

DOOMED ARCHIVE

Team Space Matrix

SLIIT Codefest 2026 Competition Submission Report

Demonstration Video

Unlisted YouTube link: <https://youtu.be/EJp3GZsvSKQ\>

Important: Replace the placeholder above with the final unlisted video link before submission.

Project Summary

DOOMED ARCHIVE is an evidence-first retrieval and reasoning system for the Ashen Era Archive. It connects rich visual answers, multi-document reasoning, and iterative investigation while keeping every answer grounded in source material.

Key Results

Measure	Result
Figure plates	6/6 passed
1B recall@10	0.905
Complete coverage@10	0.714
Development run	62/100 diagnostic score

Problem Statement

The archive describes an invented world in which hallucination is especially risky. It contains near-identical names, contradictory sources, scans, tables, and image-only answers. A useful system must therefore ground each response by subject, source, authority, and visual evidence.

Connected Sub-tracks

- 1B provides the evidence spine through multi-document retrieval.
- 1C provides the search and sufficiency loop for iterative investigation.
- 1A provides the rich renderer for visual, cited answers.

All three sub-tracks use one shared engine rather than separate prototypes.

Solution Overview

- Ingest PDF, DOCX, Markdown, text, tables, scans, and images while preserving page, section, content type, and source tier.
- Index 2,474 chunks with local dense embeddings, BM25 lexical search, and Reciprocal Rank Fusion.
- Describe 70 images with a vision model and retain label-value bindings.
- Use a sourced entity graph and a bounded A1-A6 reasoning state machine.

System Architecture

The system has a hard boundary at POST /v1/search. P1 produces evidence; P2 reasons only over the returned evidence. **Processing Flow**

- Corpus: 341 read-only files
- Ingestion: 236 documents and 3,134 blocks
- Vision and graph: 70 VLM-described images and 198 entities
- Index: BM25 plus Qdrant over 2,474 chunks
- Search API: POST /v1/search
- A1 Analyze and Link
- A2 Retrieve
- A3 Sufficiency Loop
- A4 Merge and Resolve Conflicts
- A5 Compose Cited Claims
- A6 Verify Entailment
- UI and API answer delivery

Reasoning Stages

Stage	Main responsibility	Output
A1	Preserve names, link entities, classify intent, and split questions.	Normalized query, intent, and entity seeds.
A2 + A3	Retrieve evidence, assess sufficiency, and repeat within a fixed budget.	Chunks, edges, missing facts, and trace.
A4	Deduplicate evidence and handle disagreement using authority tiers.	Evidence bundle, conflicts, and reliability notes.
A5 + A6	Create discrete cited claims, select assets, preserve provenance, and verify entailment.	Verified answer packet.

Answer Contract

Each answer packet include claims, per-claim citation identifiers, excerpts and hashes, figures, conflicts, missing information, warnings, a source map, an execution trace, model usage, latency, and cache information.

Important Technical Decisions

One engine across all sub-tracks

1A, 1B, and 1C share ingestion, retrieval, reasoning, verification, and observability.

Direct orchestration

The system does not depend on LangChain or LlamaIndex; the reasoning stages are implemented directly.

Vision descriptions instead of OCR alone

For Emberdeep, vision recovered values 800, 2400, and 6000 while the requested value was 1114. Tesseract found labels but not the required digits.

Directory-first authority tiers

This produced zero unclassified sources, compared with 87 under filename-only classification. Lower-tier evidence is marked as preferred or left unresolved when appropriate. **Hybrid retrieval**

Local BGE-small dense embeddings are combined with BM25 through Reciprocal Rank Fusion.

Intent-aware graph expansion

Graph expansion improved complete coverage from 0.429 to 0.714. Reranking helped 1A but reduced 1B performance to 0.143, so it is applied selectively.

450-token chunks

A 600-token configuration partially embedded 8.8% of chunks. The selected size followed a measured sweep.

Evidence before generation

Prompts use clear evidence delimiters, reject unknown identifiers, quote source spans, and allow A6 to remove unsupported claims.

Approaches Tried That Did Not Work

- OCR-only extraction passed 3 of 11 checks and missed the Emberdeep numbers.
- Dense-only retrieval achieved 0.000 complete coverage.
- Applying reranking everywhere added about 2.4 seconds and reduced 1B coverage from 0.429 to 0.143.
- Section expansion reduced coverage from 0.429 to 0.286.
- Adding more graph edges reduced coverage from 0.714 to 0.571.
- The 600-token configuration caused partial embedding and truncation.

Verified Status and Evaluation

The latest answer evaluation used 20 development questions and 8 unanswerable questions. Results were human rated against the team's own gold answers. The approximately 62/100 result is a development diagnostic, not an official score.

Evaluation group	N	Clean	Borderline	Fail	Mean
1A figure plates	6	6	0	0	92
1A portraits, heraldry, relic	5	2	1	2	55
1B multi-hop	7	3	1	3	53
1C contradiction	2	0	1	1	38
20-question development set	20	11	3	6	62
Unanswerable set	8	4	2	2	64

What Works

- All six figure-plate checks passed.
- Hybrid retrieval with graph expansion achieved 0.905 recall@10 and 0.714 complete coverage@10.
- Evidence controls keep generation tied to retrieved sources.
- Six of eight unanswerable questions avoided unsupported assertions.

What Is Not Yet Reliable

- Multi-hop composition failed 3 of 7 development checks.
- Contradiction answers sometimes selected the wrong source or omitted competing evidence.
- Descriptive image answers can depend on exact asset identifiers and wording.
- Two long investigations reached the 90-second time limit.

Known Limitations

- The evaluation set is small and team-authored; there is no official gold benchmark.
- Conflict-detection recall is unknown and only two contradiction values were tested.
- Deep Semantic Search currently changes the retrieval budget from 6 to 12; it does not use a separate algorithm.

- Confidence calibration is not complete.
- The latest 28-question evaluation used GPT-4o directly. An outage caused 11 reruns, and there wasn't a cross-provider fallback yet.

AI Usage Disclosure

AI was used both inside the product and as a development collaborator. Our Team Space Matrix remains responsible for the design, source selection, testing, review, and submitted work.

AI Used Inside the Prototype

- A vision-language model generated descriptions for 70 images, including subject, objects, prose, and label-value information.
- An LLM proposed candidate relations constrained to known entities and checked by five validators.
- LLMs assist A1, A3, A5, and A6 but are not permitted to introduce outside knowledge.
- Calls are cached, retried with backoff, traced, and checked against sources. The latest evaluation used GPT-4o; earlier cached vision work used MiniMax.

AI Used During Development

- Codex and Claude tooling helped inspect the repository, propose changes, draft code and documentation, and debug failures.
- The A1-A6 skills act as specifications for prompts, schemas, validators, stopping conditions, and tests.
- Architecture decision records capture context, alternatives, results, and draft markers.
- Suggestions were retained only after tests, traces, source inspection, or evaluation. A confidence-prompt change was reverted; and extra graph edges were excluded after regression.

Disclosure Boundaries

AI-assisted work	Human-owned controls
Image descriptions and candidate graph relations	Gold answers, source and tier policy, validators, and acceptance decisions
Drafted code, tests, documentation, and debugging suggestions	Ownership, review, executed commands, commits, and final wording
Answer composition and entailment checking	Evidence prompts, citations, verification budgets, warnings, and evaluation

Evidence and Reproducibility

The repository includes reproduction instructions, schemas, A1-A6 skills, decision records, ablations, OCR-versus-VLM results, the chunk-size sweep, a Postman collection, traces, and the 9 September evaluation. No exported AI chat transcript is included, and this report does not claim that one is present.

Team Members and Contributions

Team member	Role and contribution
M.I.M Ithisham	Video scripting, development, reasoning, and orchestration.
M.A.M Eyaas	Architecture design, initialization, development, and knowledge layer.
Brammawaxanan S	UI handling, video editing, and development.
Imaadh Ifham	Testing and agent evaluation.